**Project Design Phase**
**Proposed Solution Template**

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| Date | 4 july 2026 |
| Team ID | |
| Project Name | **Credit Card Approval Prediction** |
| Maximum Marks | 2 Marks |

**Proposed Solution Template:**

| S.No. | Parameter | Description |
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| 1. | Problem Statement (Problem to be solved) | Financial institutions process a large number of credit card applications and transactions every day. Manual verification is time-consuming, prone to human errors, and may lead to incorrect approval or rejection decisions. An automated system is required to improve accuracy, reduce fraud, and speed up the decision-making process. |
| 2. | Idea / Solution description | Develop a Machine Learning-based Credit Card Approval Prediction System that preprocesses transaction data, applies feature scaling, and uses a trained Random Forest model to predict whether a transaction is Genuine or Fraudulent. The solution is deployed through a Flask web application with a simple HTML and CSS interface for easy user interaction. |
| 3. | Novelty / Uniqueness | The proposed solution combines data preprocessing, machine learning, and a user-friendly web application to provide quick and accurate predictions. It compares multiple classification models and deploys the best-performing model, making the prediction process efficient and reliable. |
| 4. | Social Impact / Customer Satisfaction | The system helps financial institutions reduce fraudulent transactions, improve operational efficiency, and support faster decision-making. Customers benefit from quicker processing and more reliable approval decisions, resulting in increased trust and satisfaction. |
| 5. | Business Model (Revenue Model) | The application can be offered as a Software-as-a-Service (SaaS) solution to banks and financial institutions through subscription-based licensing. Organizations can also integrate the prediction system into their existing banking platforms to improve fraud detection and decision-making. |
| 6. | Scalability of the Solution | The modular architecture allows future enhancements such as cloud deployment, database integration, REST APIs, real-time transaction monitoring, mobile applications, |

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| | | and support for additional machine learning or deep learning models without significant changes to the existing system. |